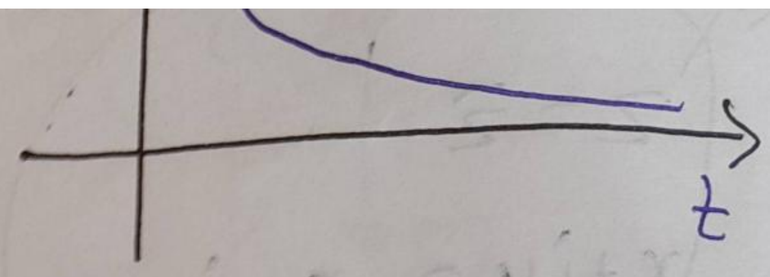

over damping

30/03/23

$$\ddot{x} + \omega_0^2 x = 0$$

$$x(t) = Ae^{\alpha t}$$

$$\alpha = \pm i\omega_0$$

$$x(t) = Ae^{i\omega_0 t} + Be^{-i\omega_0 t}$$

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$A^* = B$, $A = |A| e^{i\phi}$

$$\boxed{x(t) = 2|A| \cos(\omega_0 t + \phi)}$$

$f = -bv = -b\dot{x}$

$$m\ddot{x} = -kx - b\dot{x} \Rightarrow \ddot{x} + \frac{b}{m}\dot{x} + \frac{k}{m}x = 0$$

$$\frac{b}{m} = \nu , \quad \frac{k}{m} = \omega_0^2$$

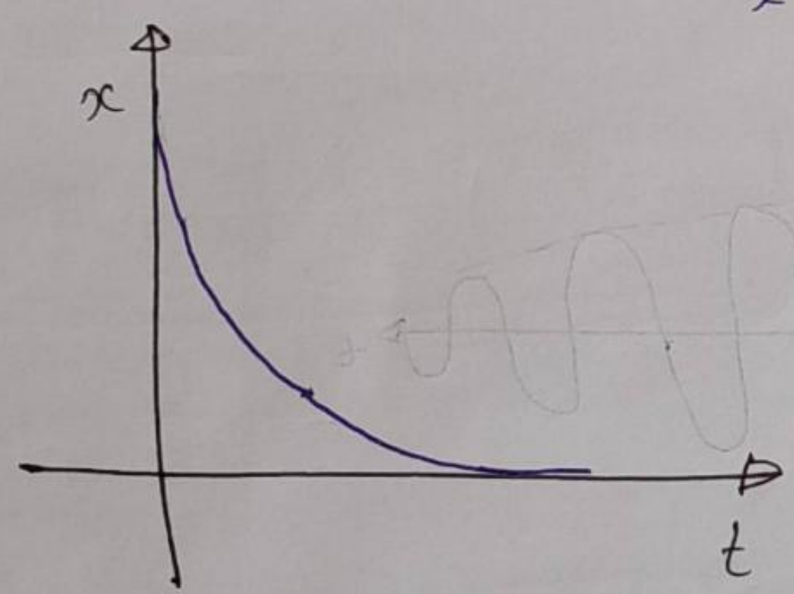
$$A = A_0 e^{\alpha t}$$

$$\alpha_\pm = \frac{\nu}{2} \pm \sqrt{\left(\frac{\nu}{2}\right)^2 - \omega_0^2}$$

$\frac{\nu}{2} > \omega_0$ $\qquad$ $\alpha_\pm < 0$

$$x(t) = Ae^{\alpha_+ t} + Be^{\alpha_- t}$$

$$= Ae^{-(\alpha_+)t} + Be^{-(\alpha_-)t}$$



$\frac{\nu}{2} < \omega_0$ $\qquad$ $\alpha_\pm = -\frac{\nu}{2} \pm i\underbrace{\sqrt{-\left(\frac{\nu}{2}\right)^2 + \omega_0^2}}_{\omega'}$

For undamped, $x(t) = 2|A| \cos(\omega t + \phi)$

$$\alpha_\pm = -\frac{\nu}{2} \pm i\omega'$$

$$x(t) = Ae^{\alpha_+ t} + Be^{\alpha_- t}$$

$$= Ae^{-\nu/2\, t + i\omega' t} + A^* e^{-\frac{\nu}{2}t - i\omega' t}$$

$$= |A| e^{-\frac{\nu}{2}t}\left[e^{i(\omega' t + \phi)} + e^{-i(\omega' t + \phi)}\right]$$

$$\boxed{x(t) = |A| e^{-\frac{\nu}{2}t}\left[2\cos(\omega' t + \phi)\right]}$$

$$\boxed{\begin{aligned} A &= |A| e^{i\phi} \\ A^* &= |A| e^{-i\phi} \end{aligned}}$$

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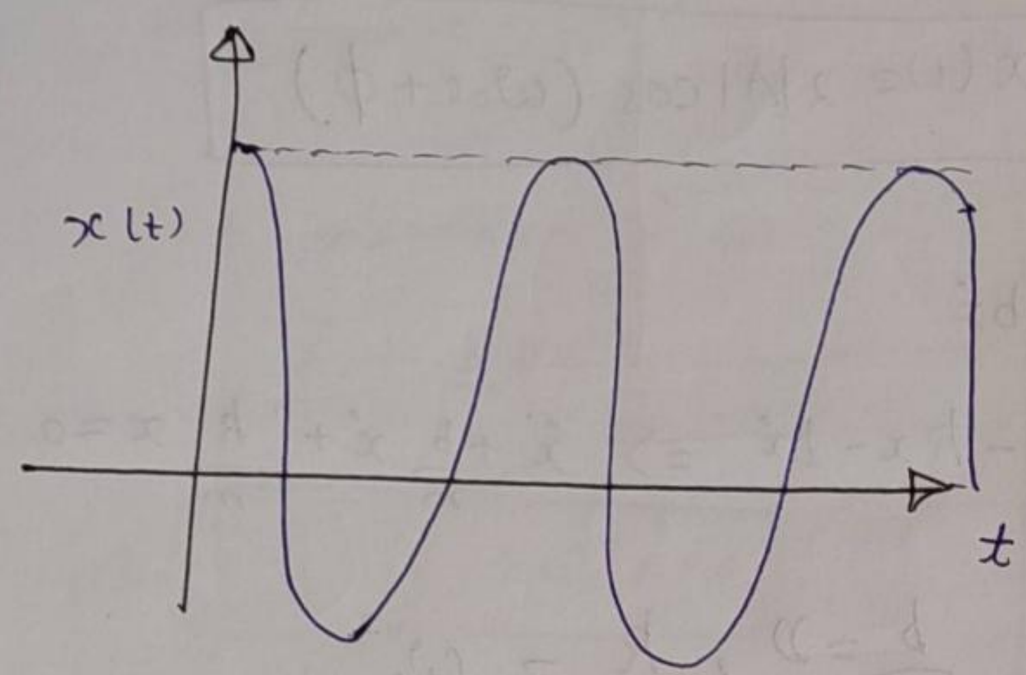
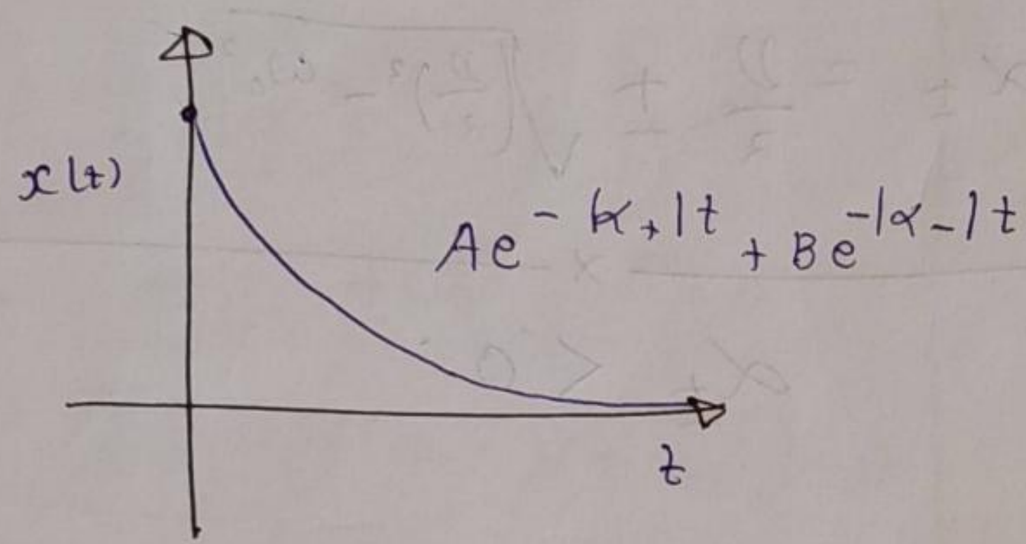
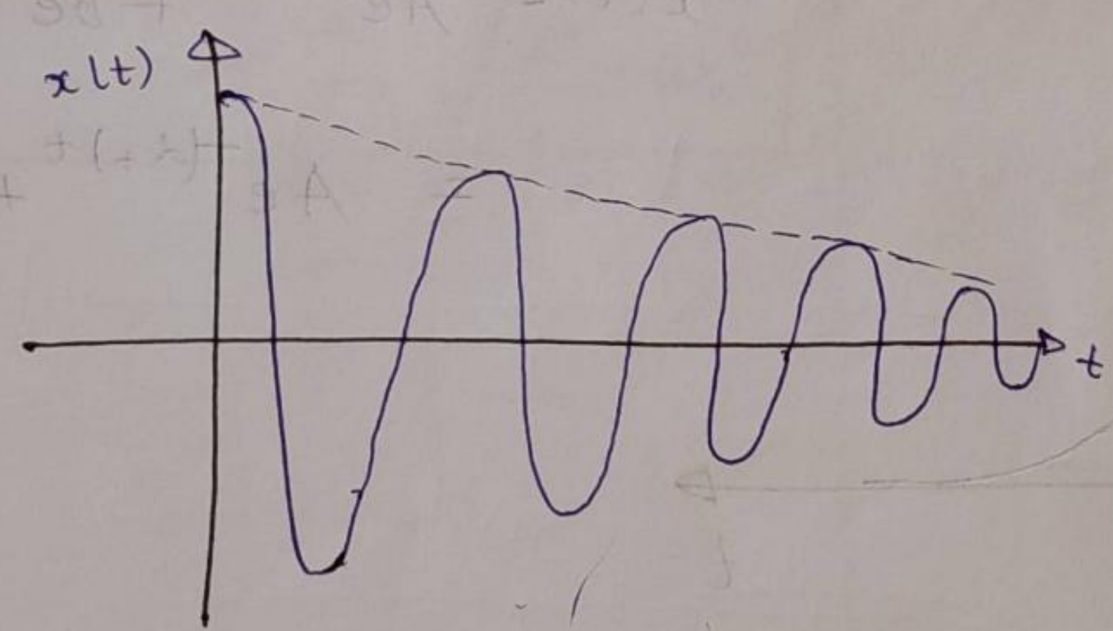
(1) Undamped, $\nu = 0$



(2) Overdamped, $\frac{\nu}{2} > \omega_0$



(3)



Quality Factor : $Q = \frac{\omega_0}{\nu}$

$$\omega' = \sqrt{\omega_0^2 - \frac{\nu^2}{4}}$$

$$\boxed{\omega' = \omega_0 \sqrt{1 - \frac{1}{4Q^2}}}$$

$$x(t) = A e^{-\frac{\nu}{2}t} \cos(\omega' t + \phi)$$
$$= A e^{-\frac{\omega_0 t}{2Q}} \cos(\omega' t + \phi)$$

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$Q >>, \quad \omega' \approx \omega_0$

$$= A e^{-\frac{\omega_0 t}{2Q}} \cos(\omega_0 t + \phi)$$

$$e^{-1} = e^{-\frac{\omega_0 t}{2Q}}$$

$$t = \frac{2Q}{\omega_0} \Rightarrow \frac{t}{T} = \frac{2Q/\omega_0}{2\pi/\omega_0} = \frac{Q}{\pi}$$

$\omega = 0$

$$A(0) = \frac{F_0/m}{\omega_0^2}$$

$\omega = \omega_0$

$$A_R(\omega_0) = \frac{F_0/m}{\nu\omega_0}$$

$$\frac{A_R(\omega_0)}{A(0)} = \frac{\omega_0}{\nu} = Q$$

$$\boxed{\frac{\nu}{2} = \omega_0}$$

critically damped

$$x(t) = A e^{-\frac{\omega_0 t}{2Q}} \cos(\omega' t + \phi)$$

$$E = \frac{1}{2} k x^2 + \frac{1}{2} m v^2$$

$$= \frac{1}{2} k \times A^2 e^{-\frac{\omega_0 t}{Q}} \cos^2(\omega' t + \phi)$$

$$+ \frac{1}{2} m \left(A^2 e^{-\frac{\omega_0 t}{Q}} \left(-\frac{\omega_0}{2Q} \cos(\omega' t + \phi) + -\sin(\omega' t + \phi) \times \omega' \right)^2 \right)$$

Forced/Driven Oscillation:

(External periodic force)

$$m\ddot{x} = -kx + (-b\dot{x}) + F_0 \cos\omega t$$

$$\ddot{x} + \underbrace{\frac{b}{m}}_{\nu} \dot{x} + \underbrace{\frac{k}{m}}_{\omega_0^2} x = \frac{F_0}{m} \cos\omega t$$

$$\boxed{\ddot{x} + \nu\dot{x} + \omega_0^2 x = \frac{F_0}{m} \cos\omega t}$$

$$\frac{\nu}{2} = \omega_0$$

$$\alpha_{\pm} = -\frac{\nu}{2}$$

$$x(t) = A e^{-\nu/2\, t} + B t e^{-\nu/2\, t}$$

Critically damped

(transient, steady) state

$$\ddot{z} + \nu\dot{z} + \omega_0^2 z = \cancel{F_0/m \cos\omega t}\ \frac{F_0}{m} e^{i\omega t}$$

$$\left\{ z = x + iy;\ z = z_0 e^{i\omega t},\ \ddot{z} = z_0 (i\omega)^2 e^{i\omega t},\ \dot{z} = z_0 i\omega e^{i\omega t} \right\}$$

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$$-\omega^2 z_0 e^{i\omega t} + \nu i\omega z_0 e^{i\omega t} + \omega_0^2 z e^{i\omega_0 t} = \frac{F_0}{m} e^{i\omega t}$$

$$z_0 e^{i\omega t}\left(-\omega^2 + i\nu\omega + \omega_0^2\right) = \frac{F_0}{m} e^{i\omega t}$$

$$\ddot{x} + \nu\dot{x} + \omega_0^2 x = \frac{F_0}{m}\cos\omega t$$

$$\ddot{z} + \nu\dot{z} + \omega_0^2 z = \frac{F_0}{m} e^{i\omega t}$$

$x \to Re(z)$

$y \to Img(z)$

$$(\ddot{x} + i\ddot{y}) + \nu(\dot{x} + i\dot{y}) + \omega_0^2 (x + iy) = \frac{F_0}{m} e^{i\omega t}$$

$$= \frac{F_0}{m}(\cos\omega t + \sin\omega t)$$

$$\ddot{x} + \nu\dot{x} + \omega_0^2 x = \frac{F_0}{m}\cos\omega t$$

$$\ddot{y} + \nu\dot{y} + \omega_0^2 y = \frac{F_0}{m}\sin\omega t$$

$$\Rightarrow \quad z_0 = \frac{F_0/m}{\underbrace{\left[(\omega_0^2 - \omega^2) + i\nu\omega\right]}_{D(\omega)}} \Rightarrow \boxed{z_0 = |z_0| e^{i\phi}}$$

$$z_0 = \frac{F_0/m\left[(\omega_0^2 - \omega^2) - i\nu\omega\right]}{\left[(\omega_0^2 - \omega^2) + \nu^2\omega^2\right]} = \frac{F_0}{m}\frac{(a - ib)}{a^2 + b^2}$$

$$z = z_0 e^{i\omega t}$$

$$z = \frac{F_0}{m}\frac{\left[(\omega_0^2 - \omega^2) - i\nu\omega\right] \times (\cos\omega t + i\sin\omega t)}{\left[(\omega_0^2 - \omega^2) + \nu^2\omega^2\right]}$$

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$$x = Re(z) = \frac{F_0}{m} \frac{\left[(\omega_0^2 - \omega^2)^2 \cos\omega t + \nu\omega \sin\omega t\right]}{\left[(\omega_0^2 - \omega^2) + \nu^2\omega^2\right]}$$

$$\Rightarrow \quad D(\omega) = (\omega_0^2 - \omega^2) + i\nu\omega = |D| e^{i\delta}$$

$$|D| = \sqrt{(\omega_0^2 - \omega^2)^2 + \nu^2\omega^2},$$

$$\tan\delta = \frac{\nu\omega}{(\omega_0^2 - \omega^2)}$$

$$\boxed{\begin{aligned} z &= x + iy \\ |z| &= \sqrt{x^2+y^2} \\ \theta &= \frac{y}{x} \end{aligned}}$$

$$z_0 = \frac{F_0/m}{D(\omega)} = \frac{F_0/m}{|D| e^{i\delta}} = \frac{F_0/m}{|D|} e^{-i\delta}$$

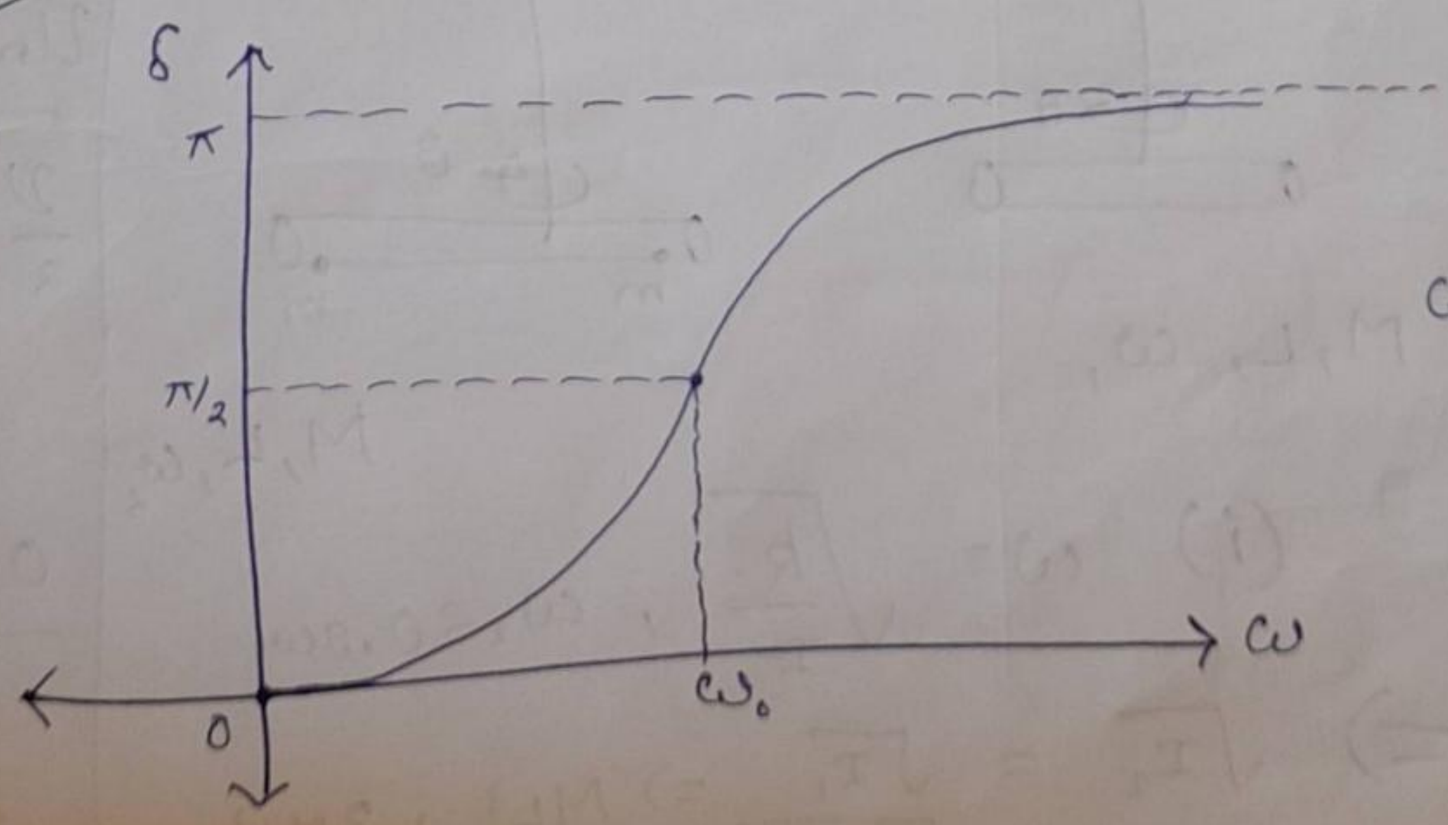
Real Part

$$\boxed{z(t) = z_0 e^{i\omega t} = \frac{F_0/m}{|D|} \cos(\omega t - \delta)}$$

Waves

Frank Crawford

02/03/23



$\omega = 0 \Rightarrow$

$$A = \frac{F_0}{m\omega_0^2} = \frac{F_0}{m\frac{k}{m}} = \frac{F_0}{k}$$

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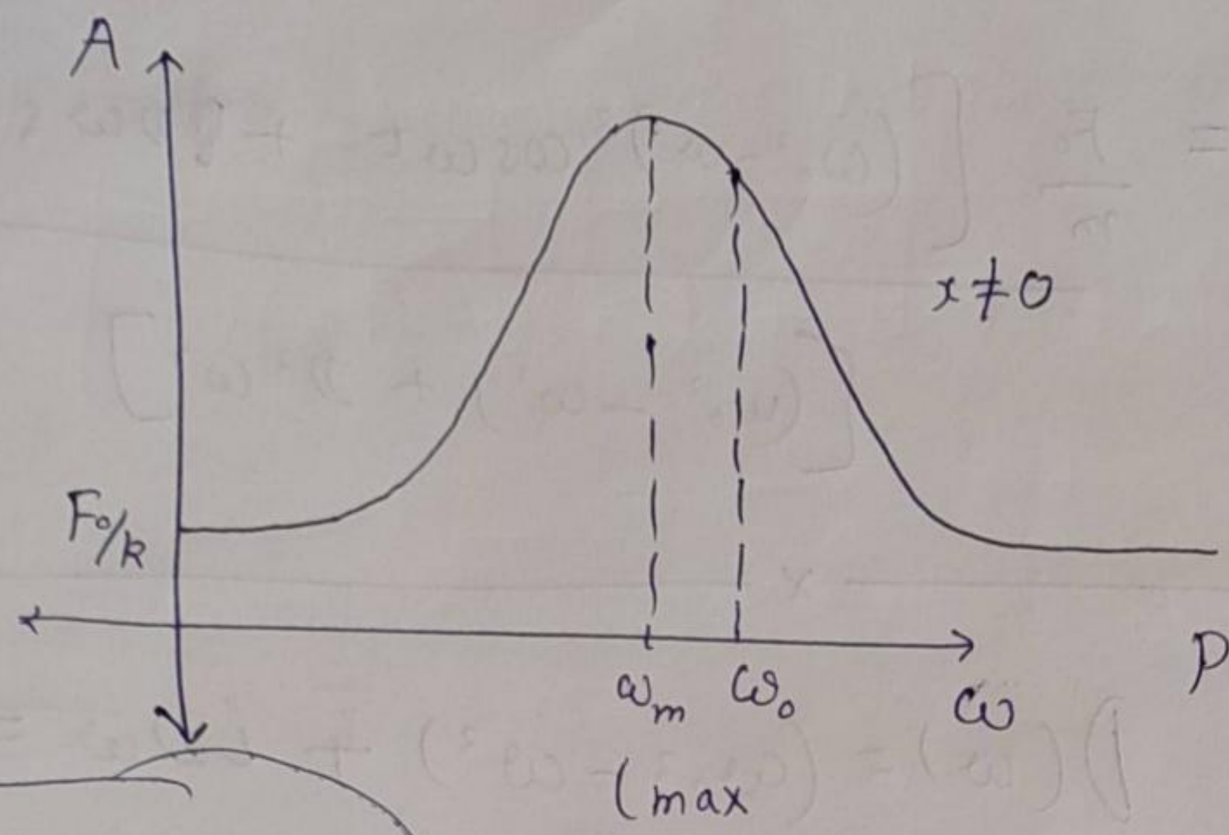


(max Amp.)

$$A = \frac{F_0/m}{\sqrt{(\omega_0^2-\omega^2)^2+\omega^2\nu^2}}$$

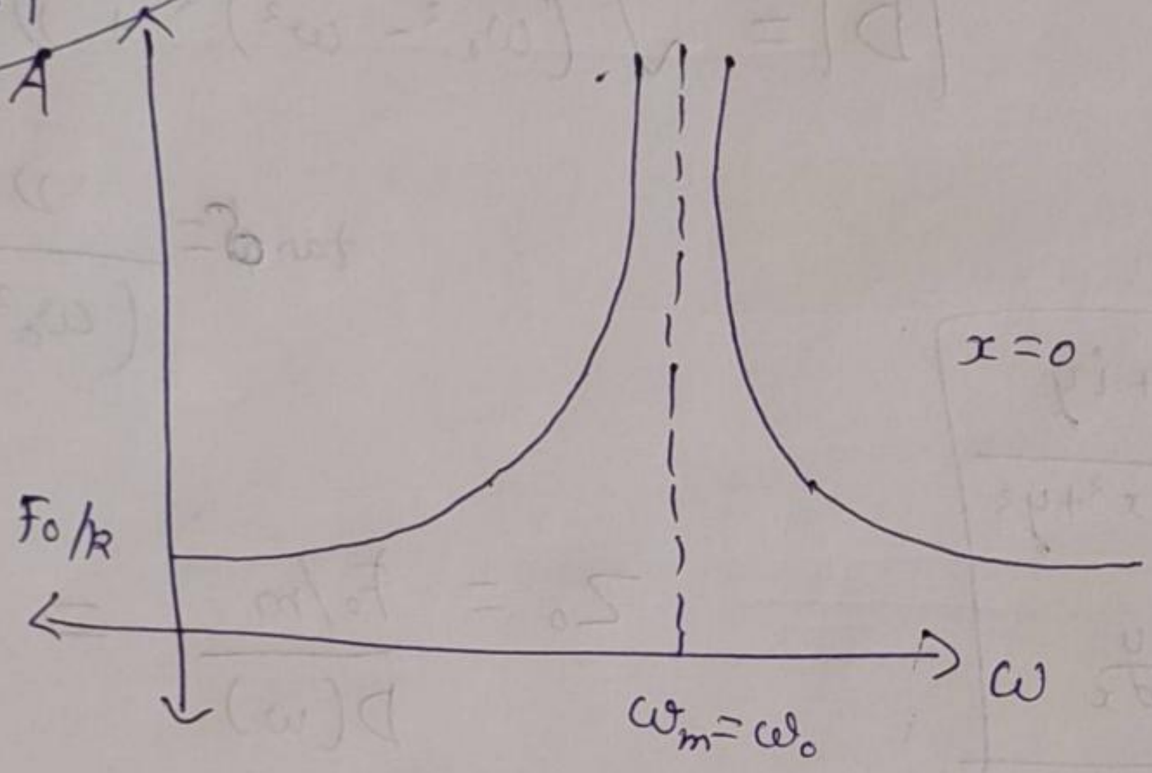
$$P = \frac{dw}{dt} = \frac{F\,dx}{dt} = Fv$$

$$P_{driving} = F_{driving}\, v$$

$$= F_0 \cos\omega t\, \dot{x}$$

$$P_{damp} = F_{damp}\, v$$

$$= -b\dot{x}\,\dot{x}$$



$\frac{F_0/m}{|D|}\cos(\omega t - \delta) + \underbrace{Ae^{-\frac{\nu t}{2}}\cos(\omega' t - \phi)}_{\text{complementary sol}^n}$

Critical damped

$\frac{\nu}{2} = \omega_0$, $(A+Bt)e^{-\frac{\nu}{2}t}$

$e^{-\omega_0 t}$; $\left(\frac{1}{\omega_0}\right)$

Under damped

$\frac{\nu}{2} < \omega_0$, $Ae^{-\nu/2\, t}\cos(\omega t +$

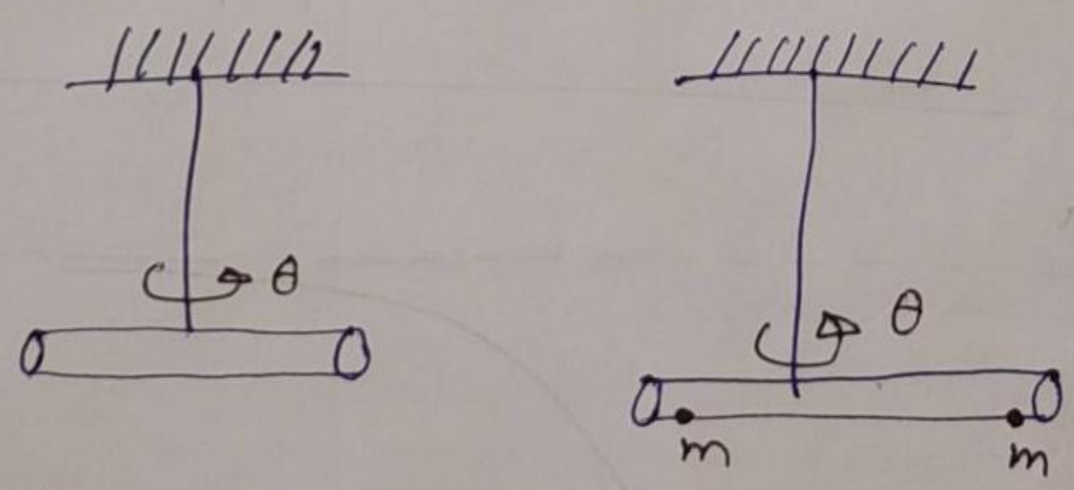
$\rightarrow \frac{2}{\nu} > \frac{1}{\omega_0}$

Over damped

$\frac{\nu}{2} > \omega_0$

$\alpha_+ = -\frac{\nu}{2} + \sqrt{\frac{\nu}{2}}$

Ex:



M, L, ω_1

M, L, ω_2

(i) $\omega = \sqrt{\frac{k}{I}}$, $\omega_2 = 0.9\omega_1$

$$\Rightarrow \sqrt{I_2} = \frac{\sqrt{I_1}}{0.9} \Rightarrow \frac{ML^2}{12} + \frac{2ML^2}{4} = \frac{1}{(0.9)^2}\left(\frac{ML^2}{12}\right)$$

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$Ae^{\alpha_1 t} + Be^{\alpha_2 t}$ $\frac{\nu}{2} > \omega_0$

$\frac{\nu}{2} < \omega_0$

$(e^{\alpha t}, te^{\alpha t})$ & $\frac{\nu}{2} = \omega_0$

$(A + Bt)e^{\alpha t}$

$$\frac{M}{12} + \frac{m}{2} = \frac{5M}{54}$$

$$\frac{\cancel{100}^{\,25}}{81 \times \cancel{12}_{\,3}}$$

$$\left\{ \left(\frac{25}{81 \times 3} - \frac{1}{12} \right) = \frac{\overset{\times 4}{25} - 81}{81 \times 3 \times 4} = \frac{19}{81 \times 3 \times 4} \right\}$$

$$\frac{m}{\cancel{2}M} = \frac{19}{81 \times 3 \times \cancel{4}_{\,2}}$$

$$\boxed{\frac{m}{M} = \frac{19}{486}} = 0.039$$

⇒ Ex: Lenard- Jones Potential ⇒

$$U(r) = 4\epsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^{6} \right]$$

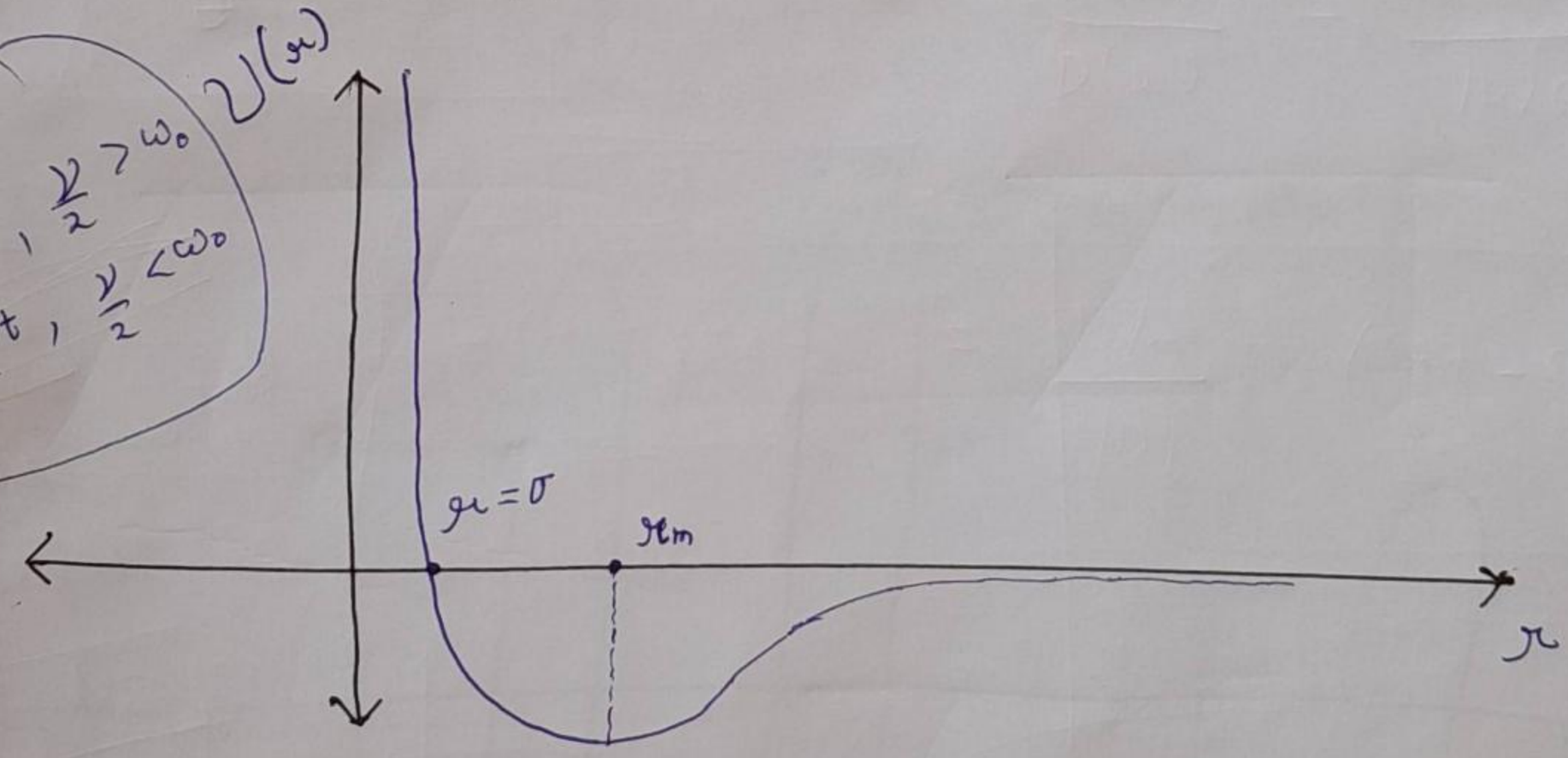
ϵ = Depth of potential, σ = size of atom (Diameter)

(2) $e^{-\omega_0 t}$

(1) $e^{-\frac{\nu}{2}t}$, $\frac{\nu}{2} > \omega_0$

(4) $e^{-\frac{\nu}{2}t}$, $\frac{\nu}{2} < \omega_0$

(3)



$\frac{\nu}{2}$

$$F = -\frac{dU}{dr} = -4\epsilon \left[12 \left(\frac{\sigma}{r} \right)^{11} \left(\frac{-\sigma}{r^2} \right) - 6 \left(\frac{\sigma}{r} \right)^{5} \left(\frac{-\sigma}{r^2} \right) \right]$$

$$\Rightarrow F = -4\epsilon \left[-12 \frac{\sigma^{12}}{r^{13}} + \frac{6\sigma^6}{r^7} \right]$$

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